Kundan Mergu
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GGIS 527
Final Project Proposal

# Analyzing Crop Production Across India

## Introduction:

India, an agrarian economy, has witnessed significant transformations in its agricultural patterns due to factors such as urbanization, climate change, and technological advances. Analyzing these shifts at the district level provides insights into food security, resource allocation, and sustainable farming practices. However, while existing studies extensively explore macro-level trends (state or national scale), granular district-level insights over extended timeframes (1997–2023) remain underexplored.

### Research Questions:

- **Spatial Patterns**: What are the dominant crops grown in different districts of India from 1997 to 2023?
- **Temporal Changes**: How have cropping patterns evolved over the 26-year period?
- **Future Projections**: How will these patterns likely change by 2030, considering historical trends and environmental factors?
- **Impact Assessment**: What are the implications of these changes on food security and agricultural sustainability?

## Literature Review:

Recent studies have explored various ML approaches for crop yield prediction in India:

1. Saravanan and Bhagavathiappan (2024): Developed models using CatBoost regression and a hybrid deep learning approach combining spatio-temporal attention-based convolutional neural networks with bidirectional long short-term memory (BiLSTM) networks, achieving an $R^2$ value of 0.99.

2. Kumar, Sharma, and Verma (2024): Implemented supervised ML algorithms—including K-nearest neighbors, logistic regression, decision trees, and random forests—to predict yields based on factors like fertilizer usage, temperature, humidity, pH, and rainfall, with the random forest model attaining 99.77% accuracy.

3. Thomas (2023): Reviewed various ML and deep learning techniques applied to Indian agriculture, highlighting the effectiveness of models that incorporate environmental parameters such as temperature, rainfall, and soil type.

4. Mahesh and Soundrapandiyan (2024): Evaluated gradient-based algorithms—CatBoost, LightGBM, and XGBoost—for crop yield prediction, with CatBoost achieving the highest precision and an accuracy rate of 99.123%.

5. De Clercq and Mahdi (2024): Investigated the feasibility of ML-based rice yield prediction at the district level using climate reanalysis data, achieving an out-of-sample $R^2$ of up to 0.82.

## Background:

Integrating machine learning algorithms, such as K-NN (K-Nearest Neighbors) and OLS (Ordinary Least Squares), into agricultural research is a field that needs to be explored a lot more. These methods can reveal more nuances in spatial relationships in crop cultivation changes over time of the crops cultivated. This in turn offers data points for generating predictive models for predicting the future of Indian agriculture. Addressing this research gap is critical for policy formulation and adaptation strategies in agriculture.

## Methods

### Data Sources

- Crop Data: District-wise crop production and yield data (1997–2023) from Food and Agriculture Organization of UN.
- Climate Data: Rainfall, temperature, and drought indices from Kaggle.

### Methods of Analysis

- Data Preprocessing:
  - Cleaning, handling missing values, and normalizing datasets.
  - Temporal alignment of district boundaries, considering administrative changes.
- Exploratory Data Analysis (EDA):
  - Temporal trends using line plots and heatmaps.
  - Regional distribution patterns using choropleth maps.
- Modeling Approaches:
  - K-NN: For clustering districts with similar cropping patterns.
  - OLS Regression: To identify relationships between crop yields and influencing factors.
  - Advanced ML Algorithms: Random Forest, Time-Series Forecasting (ARIMA/LSTM), and Support Vector Machines for predictive modeling.

- Validation:
    - Cross-validation techniques to ensure model robustness.
    - Comparison of ML model predictions with historical trends.

**Limitations:**

- Reliance on older data in districts with incomplete records.
- Data inconsistencies due to administrative boundary changes might be an issue, but there have been rare cases of boundary changes so it might not have an effect.
- No active data on agrarian policies and policy changes in regards to subsidies and programs to

**Delimiters:**

- Spatial scope is restricted to Indian districts for operational feasibility.
- The time interval is fixed at 1997–2023 to ensure dataset consistency.

## Expected Results:

**Spatiotemporal Trends:**

- Identification of regions with significant crop transitions (e.g., shift from subsistence to cash crops).
- Hotspots of monocropping or crop diversification.

**Predictive Outputs:**

- Future crop distribution maps under different scenarios.
- Key driving factors influencing specific crop changes (e.g., rainfall for paddy).

**Visualizations:**

- Choropleth maps of crop areas by year.
- Time-series plots for district-level crop yields.
- Factor importance plots for ML models.

## Hyper-Parameters and Performance Metrics:

Hyper-Parameters

1. **KMeans Clustering**
   - **n_clusters**: 5 (used to group districts with similar crop production patterns).
   - **random_state**: 42 (ensures reproducibility).
   - **PCA (Principal Component Analysis)**:
     - **n_components**: 2 (for dimensionality reduction to visualize clusters).
2. **LSTM Model for Time-Series Forecasting**
   - **LSTM Layers**:
     - Layer 1: 64 units, `activation='relu'`, `return_sequences=True`.
     - Layer 2: 32 units, `activation='relu'`.
   - **Dropout Rate**: 0.2 (regularization to prevent overfitting).
   - **Dense Layers**:
     - Layer 1: 16 units, ReLU activation.
     - Output Layer: Single unit (for regression output).
   - **Optimizer**: Adam (adaptive learning rate optimization).
   - **Loss Function**: Mean Squared Error (MSE).
   - **Batch Size**: 32.
   - **Epochs**: 100 (number of full passes through the dataset).
   - **Window Size**: 5 (number of past timesteps used for prediction).
3. **Data Preprocessing**
   - **StandardScaler**: Scales input features to zero mean and unit variance.
   - **MinMaxScaler**: Scales input features to a range of [0, 1].

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Performance Metrics:

1. **Clustering Performance**
   - **Visual Interpretation**: Scatter plots with PCA dimensions visually analyze cluster separations.
   - **Cluster Analysis**: Summarizes dominant crops and characteristics for each cluster.
2. **Regression and Forecasting Metrics**
   - **Mean Squared Error (MSE)**: Measures the average squared error between predictions and actual values. Lower MSE indicates better model performance.
   - **Mean Absolute Error (MAE)**: Represents the average magnitude of prediction errors.
   - **Validation Loss**: Tracks the error on the validation dataset, crucial for detecting overfitting.
3. **Correlation Analysis**
   - **Pearson Correlation Coefficient**: Measures the linear relationship between climate variables and crop production. For example:
     - Monsoon rainfall: Positive correlation with production (0.305).
     - Pre-monsoon rainfall: Negative correlation (-0.379).
     - Annual temperature: Near-zero correlation (-0.017).

4. **Time-Series Forecasting**
    - **Training and Validation Loss Curves**: Used to monitor convergence during training.
    - **Forecast Evaluation**:
        - Visualization of historical vs. forecasted production.
        - Confidence intervals indicate the reliability of predictions.

## Discussion and Conclusion:

This analysis highlights India's evolving agricultural landscape at a district level, emphasizing the role of climatic factors in shaping crop production patterns. Predictions suggest an increasing focus on cash crops in specific regions, while others may face challenges due to climatic variability.

**Conclusions:**

1. Rainfall and temperature significantly influence crop yields.
2. Monocropping trends in certain areas could threaten sustainability.
3. Future projections offer actionable insights for policymakers to ensure food security.

The study demonstrates the utility of machine learning in agricultural analysis and paves the way for further district-level research to guide adaptive strategies.

## Citation:

Food and Agriculture Organization of the United Nations. *Crops and livestock products (1997–2023).* Accessed November 16th, 2024. https://www.fao.org/faostat/en/#data/QCL

Kaggle. *Indian Agriculture and Climate Dataset (1961-2018).* Accessed November 16th, 2024. https://www.kaggle.com/datasets/swarooprangle/indian-agriculture-and-climate-dataset-1961-2018.

Saravanan, K., & Bhagavathiappan, A. (2024). Predicting Crop Yields Using Hybrid Deep Learning Models in Indian Agriculture. *SpringerLink*. Retrieved from https://link.springer.com/article/10.1007/s11600-024-01312-8

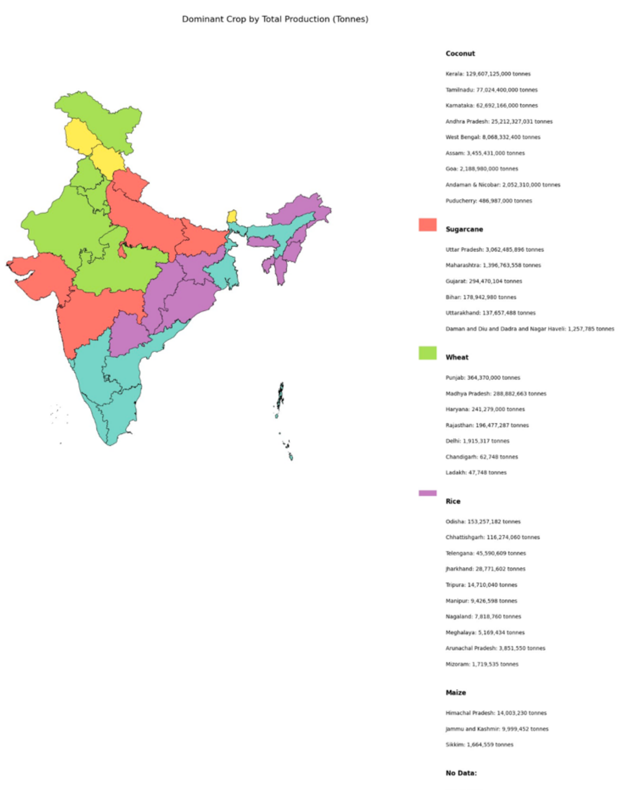
Kumar, S., Sharma, R., & Verma, A. (2024). Machine Learning Approaches for Crop Yield Prediction in Indian Districts. *SpringerLink*. Retrieved from https://link.springer.com/chapter/10.1007/978-981-99-8135-9_18

Thomas, P. (2023). A Review of Machine Learning Models in Indian Agriculture. *American Institute of Physics (AIP) Conference Proceedings*. Retrieved from https://pubs.aip.org/aip/acp/article/2773/1/020002/2892481

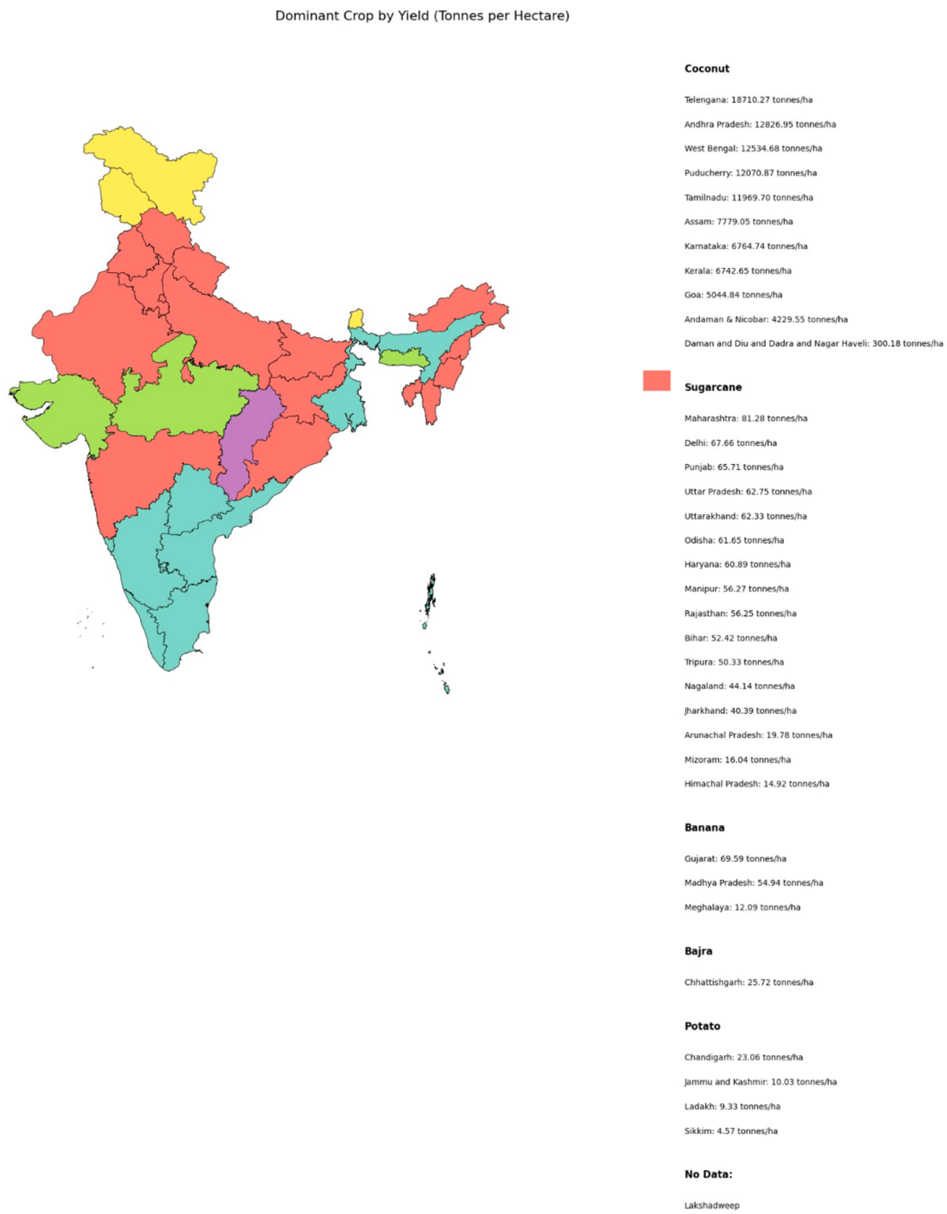
Mahesh, S., & Soundrapandiyan, A. (2024). Evaluating Gradient-Boosting Algorithms for Crop Yield Prediction. *PLoS ONE*. Retrieved from https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0291928

De Clercq, M., & Mahdi, A. (2024). Machine Learning for Rice Yield Prediction at District Level. *arXiv Preprint*. Retrieved from https://arxiv.org/abs/2403.07967

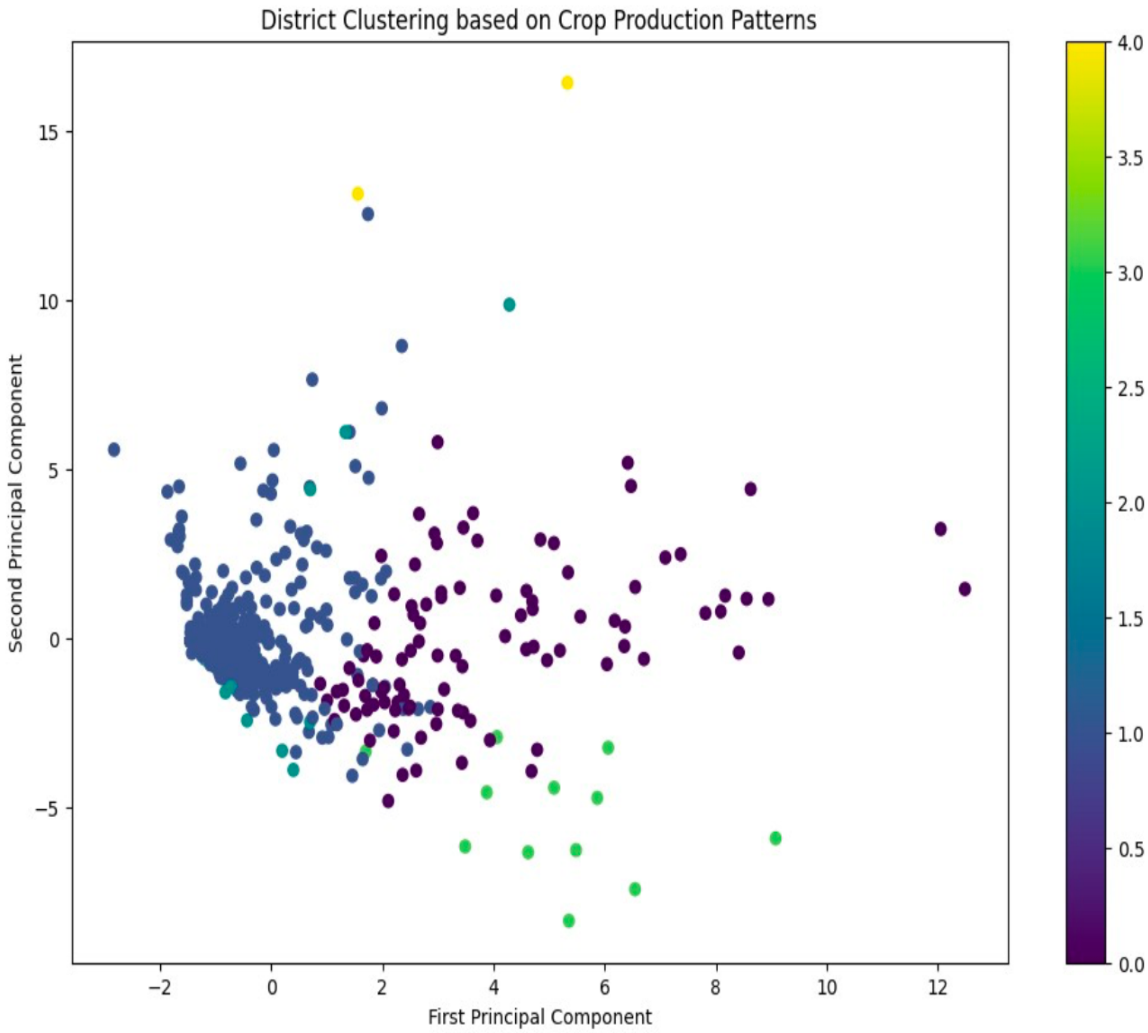

## Figures and Tables:



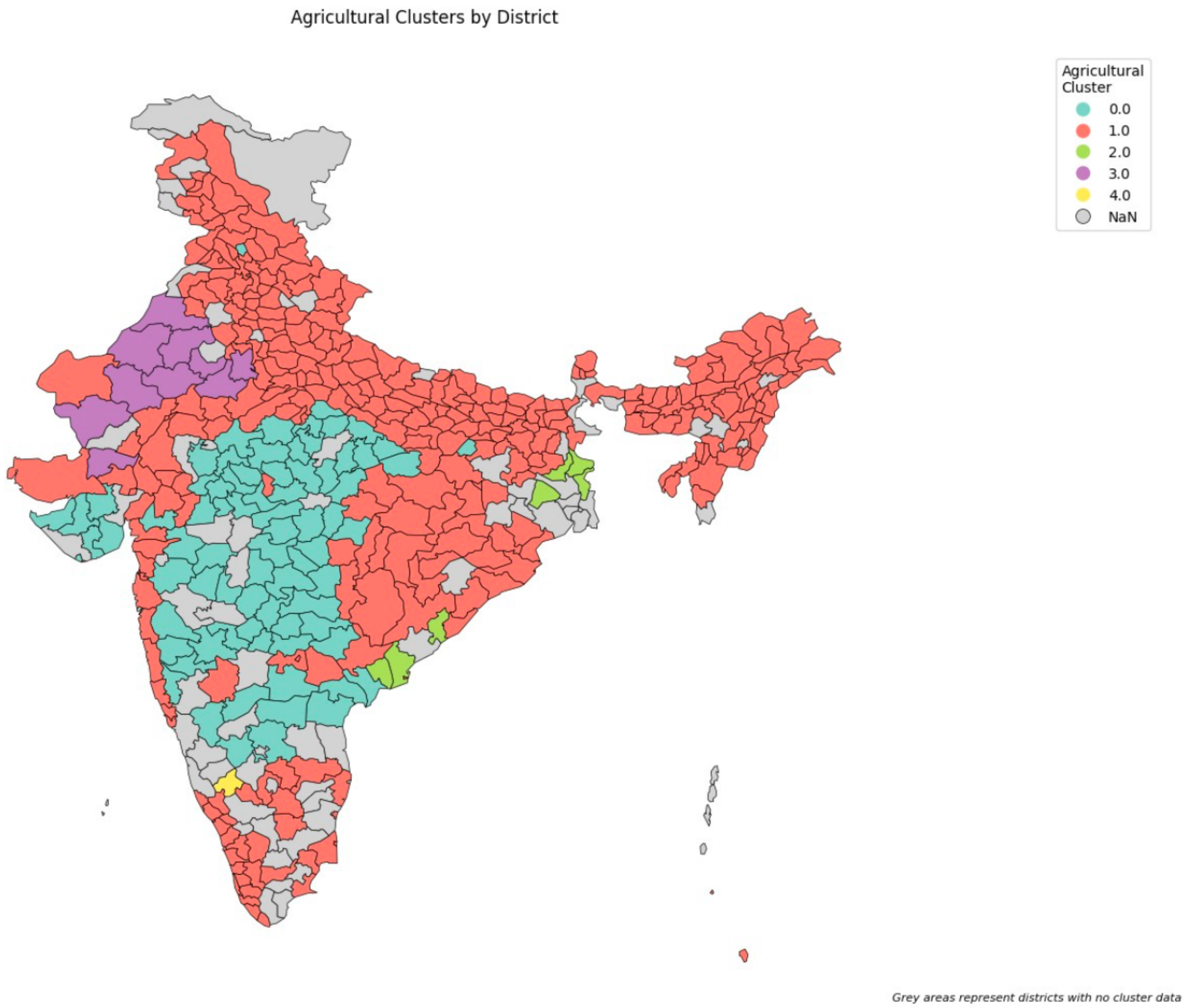

This image depicts a choropleth map highlighting the dominant crop production across various states in India, categorized by total production in tons. Each state is color-coded based on its leading crop, with a legend on the right providing detailed information about the specific crop and its corresponding production volume in tons.



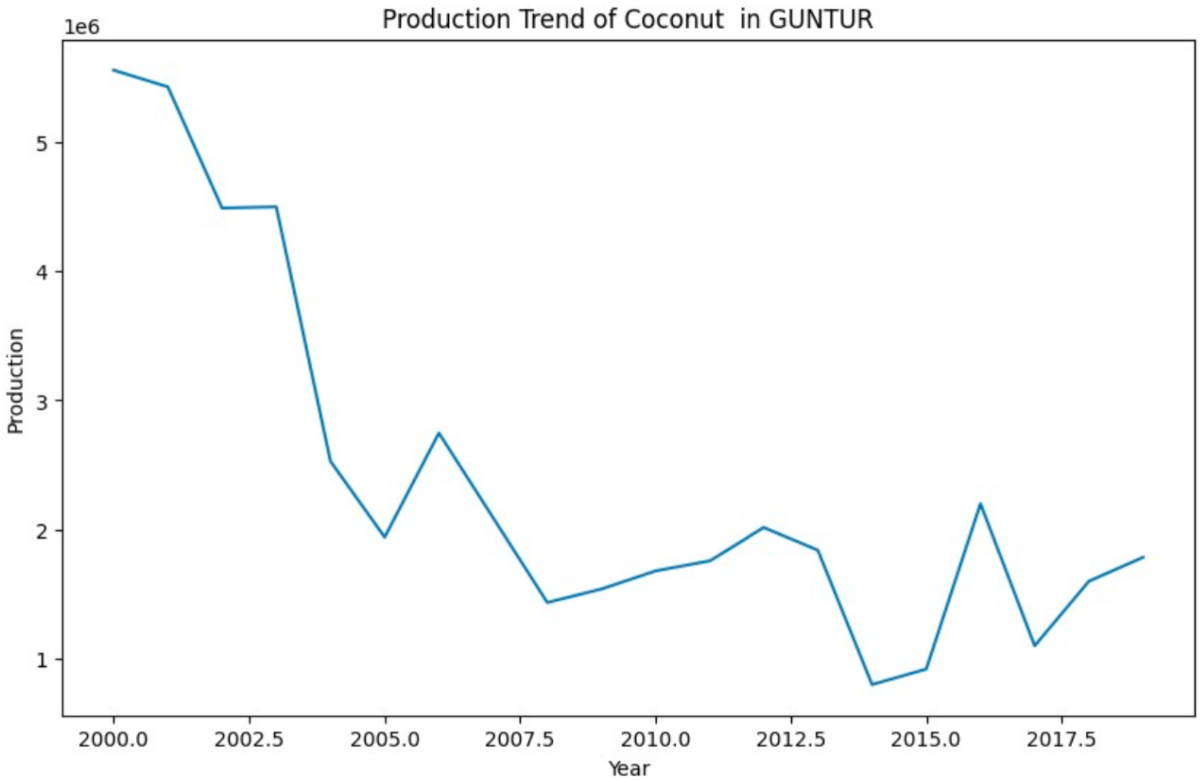

This map illustrates the dominant crops in India based on yield (measured in tons per hectare) across various states. Each state is color-coded according to the crop with the highest yield, as indicated in the legend. The accompanying legend provides detailed information on the specific crops and their yield values, highlighting regional agricultural efficiencies and productivity.

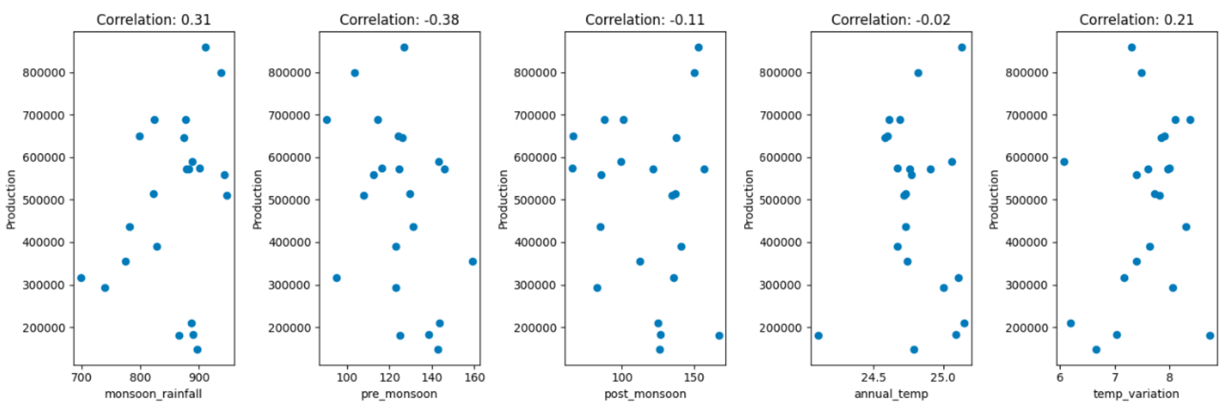



This scatter plot represents district-level clustering based on crop production patterns, visualized through principal component analysis (PCA). Each point corresponds to a district, with its position determined by two principal components capturing variations in crop production data. The color scale indicates cluster membership, revealing groupings of districts with similar agricultural profiles, while outliers suggest unique cropping patterns.

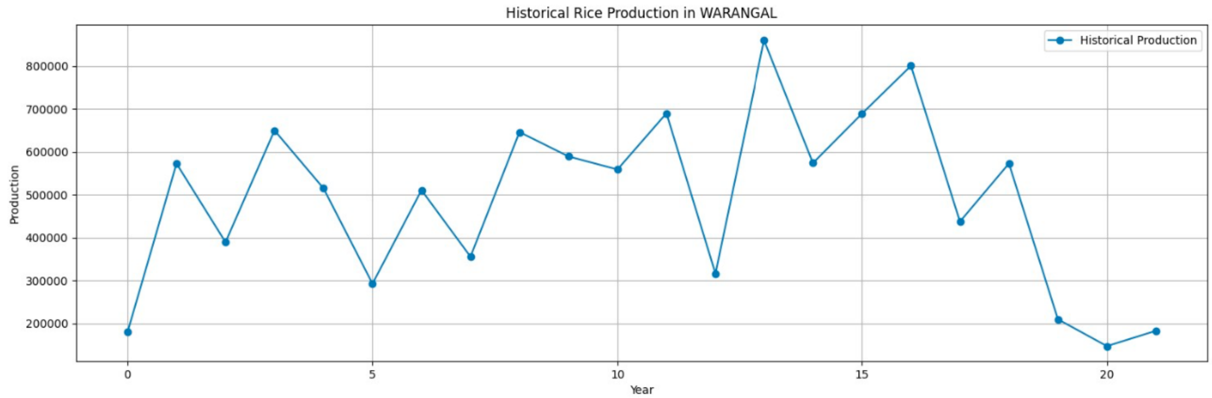


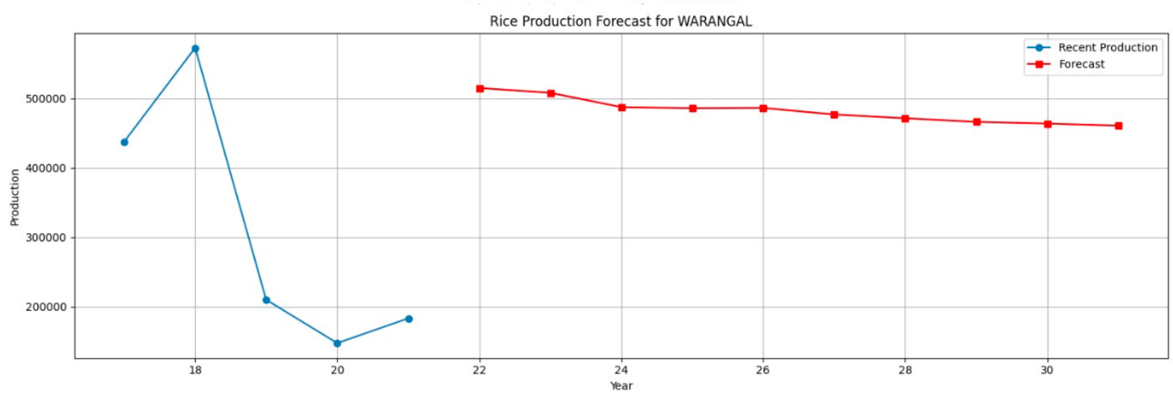

This map displays agricultural clusters across Indian districts, where each cluster is represented by a unique color. These clusters, derived from crop production patterns, group districts with similar agricultural characteristics. Grey areas indicate districts with no cluster data available.




The above visualization presents the production trend of coconut in Guntur district over time (2000–2018), showing a significant decline followed by stabilization and fluctuations in recent years. Below the trend graph are scatter plots illustrating the correlation between coconut production and key climatic factors: monsoon rainfall, pre-monsoon rainfall, post-monsoon rainfall, annual temperature, and temperature variation. The correlation values highlight varying degrees of influence, with monsoon rainfall (0.31) and pre-monsoon rainfall (-0.38) showing notable impacts on production.




This visualization includes two graphs depicting rice production in Warangal. The top graph showcases the historical trends in rice production, highlighting fluctuations over time due to varying environmental and socio-economic factors. The bottom graph provides a forecast for rice production in the region, indicating a slight decline in the upcoming years based on recent trends and predictive modeling.